

NLP Experiment 1

Experiment-1.1: Read the employee dataset and display the average salary of each department using a bar chart.

```
import pandas as pd
import matplotlib.pyplot as plt
from google.colab import drive

drive.mount('/content/drive')
```

Mounted at /content/drive

```
df = pd.read_csv('/content/drive/MyDrive/Copy of employee_information_100.csv')

print(df.head())
```

	Employee_ID	Name	Department	Age	Gender	Salary	Experience_Years
0	E001	Diana1	HR	39	Male	58000	8
1	E002	Diana2	Operations	56	Male	105000	27
2	E003	Bob3	HR	27	Male	59000	32
3	E004	Neha4	HR	57	Male	113000	34
4	E005	Nisha5	IT	50	Female	30000	10

Start coding or [generate](#) with AI.

```
print(df.columns)
```

```
Index(['Employee_ID', 'Name', 'Department', 'Age', 'Gender', 'Salary',
      'Experience_Years'],
      dtype='object')
```

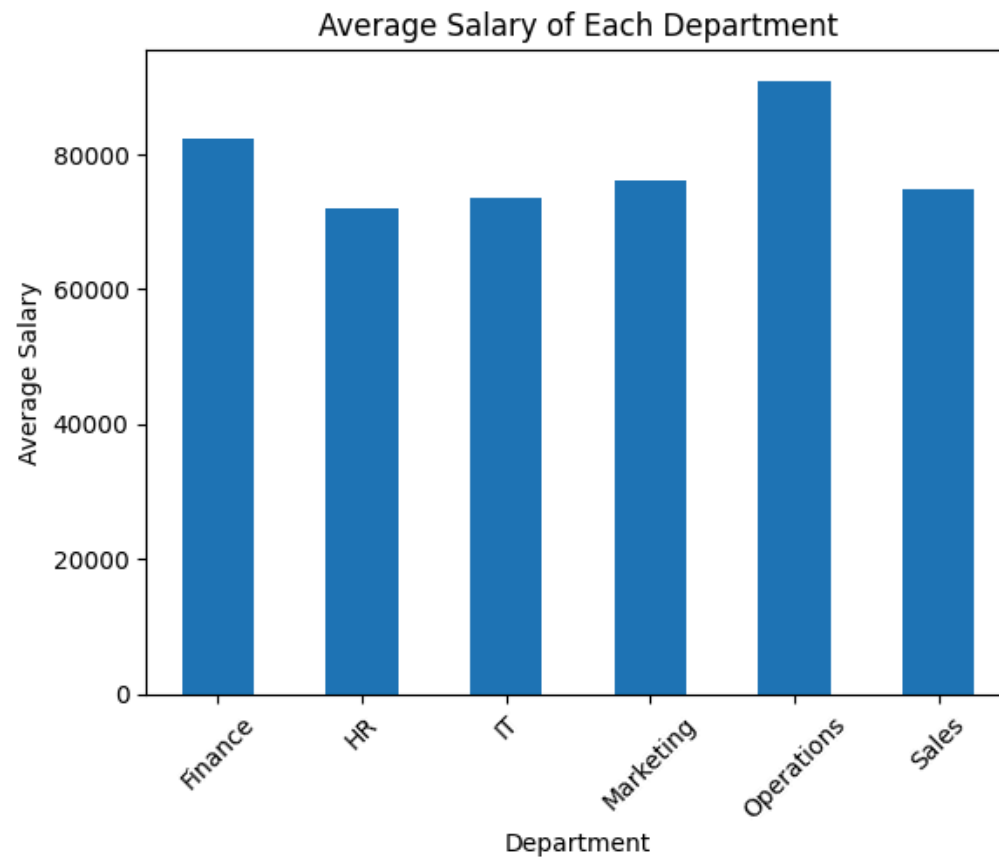
```
avg_salary = df.groupby('Department')['Salary'].mean()

print(avg_salary)
```

Department	
Finance	82416.666667
HR	72095.238095
IT	73611.111111
Marketing	76100.000000

```
Operations    90875.000000  
Sales        74769.230769  
Name: Salary, dtype: float64
```

```
avg_salary.plot(kind='bar')  
  
plt.title('Average Salary of Each Department')  
plt.xlabel('Department')  
plt.ylabel('Average Salary')  
plt.xticks(rotation=45)  
plt.show()
```



Experiment 1.2: Display the number of employees working in each department.

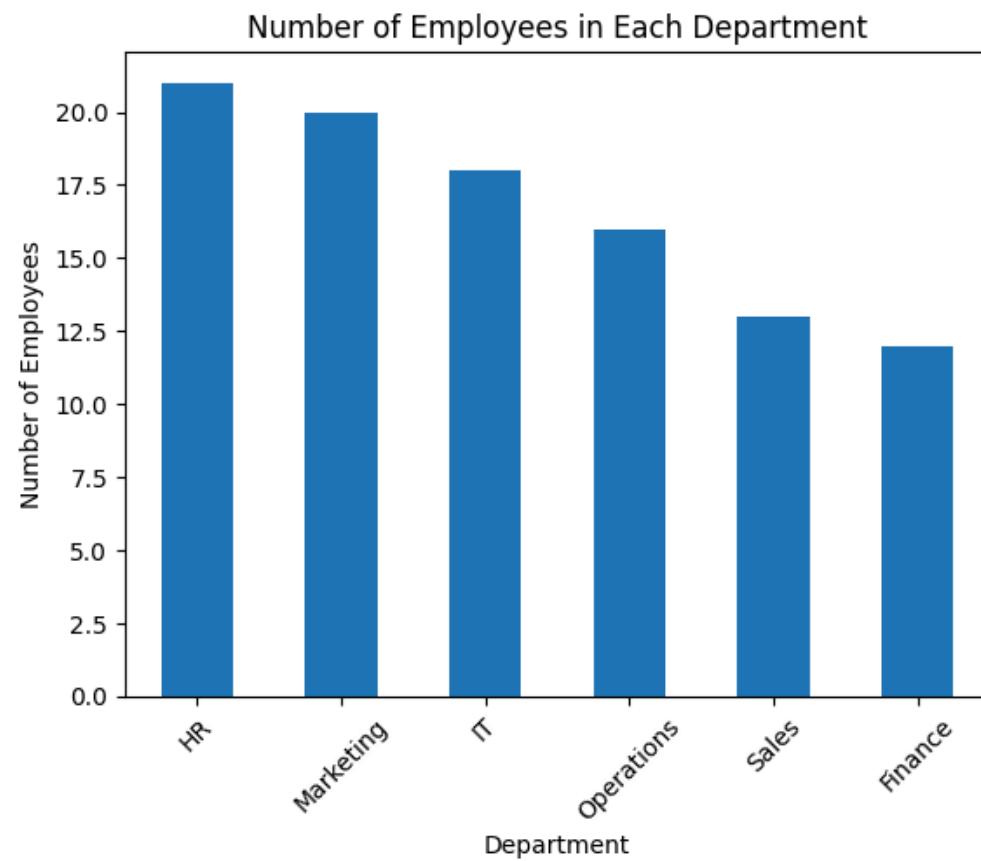
```
employee_count = df['Department'].value_counts()
```

```
print(employee_count)
```

```
Department  
HR          21  
Marketing   20  
IT          18  
Operations  16  
Sales       13  
Finance     12  
Name: count, dtype: int64
```

```
employee_count.plot(kind='bar')
```

```
plt.title('Number of Employees in Each Department')  
plt.xlabel('Department')  
plt.ylabel('Number of Employees')  
plt.xticks(rotation=45)  
plt.show()
```



Experiment 1.3: Display the percentage of male and female employees using a pie chart

```
gender_count = df['Gender'].value_counts()

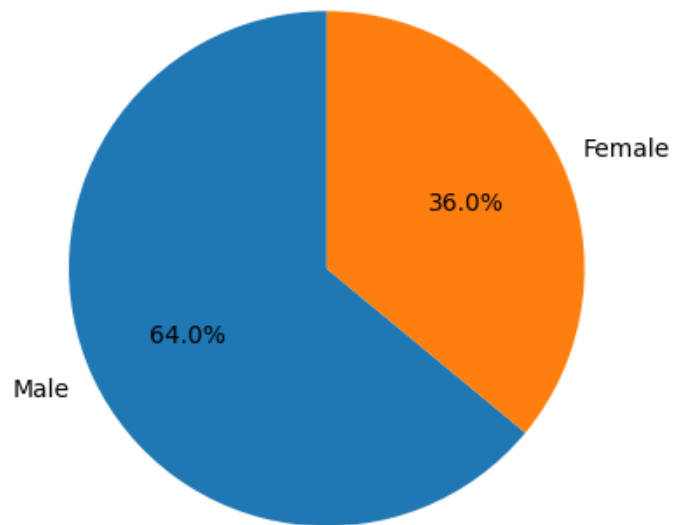
print(gender_count)

gender_count.plot(
    kind='pie',
    autopct='%1.1f%%',
    startangle=90
)

plt.title('Percentage of Male and Female Employees')
plt.ylabel('')
plt.show()
```

```
Gender
Male      64
Female    36
Name: count, dtype: int64
```

Percentage of Male and Female Employees

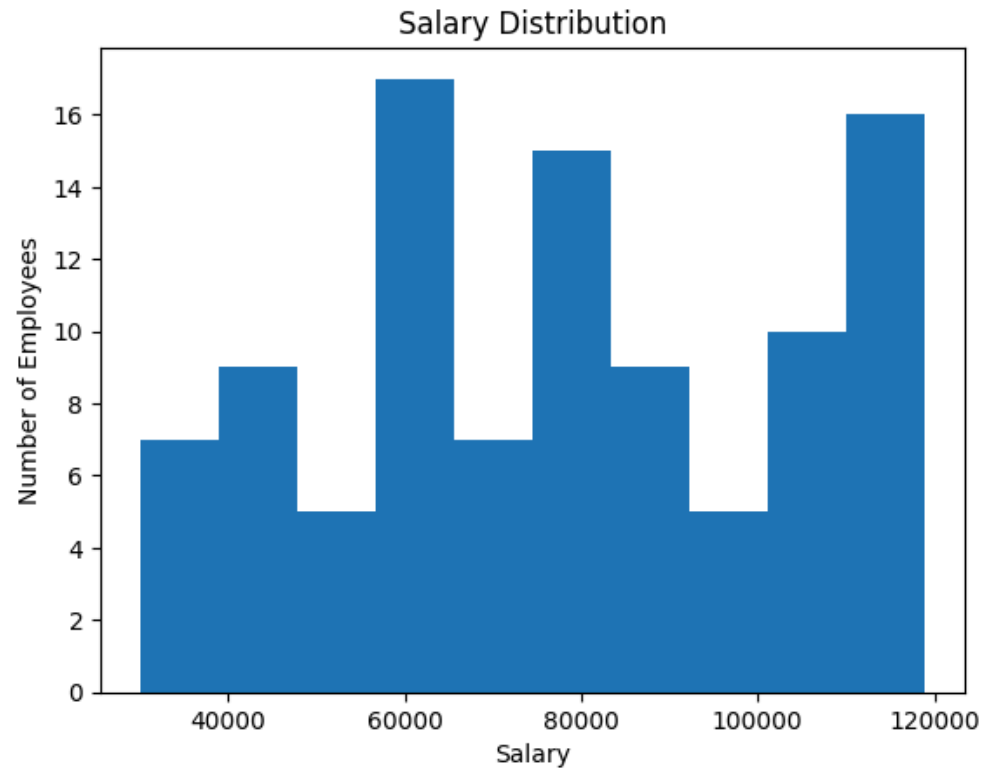


Experiment 1.4: Draw a histogram showing salary distribution.

```
plt.hist(df['Salary'], bins=10)

plt.title('Salary Distribution')
plt.xlabel('Salary')
plt.ylabel('Number of Employees')

plt.show()
```

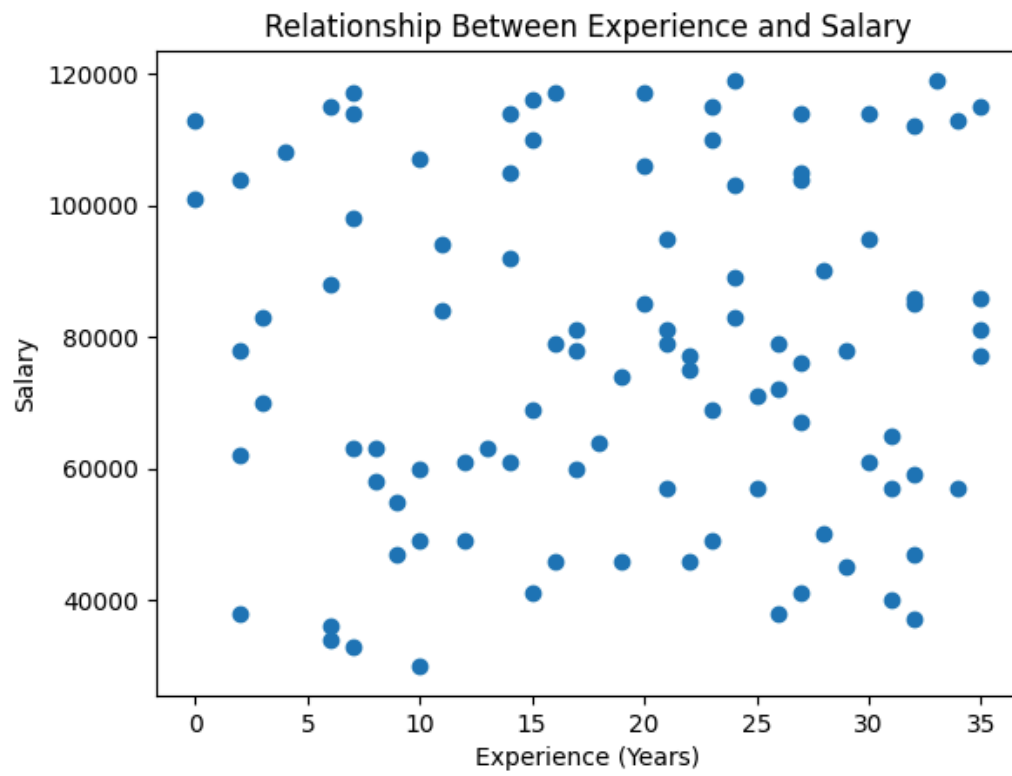


Experiment 1.5: Draw a scatter plot showing the relationship between experience and salary.

```
plt.scatter(df['Experience_Years'], df['Salary'])

plt.title('Relationship Between Experience and Salary')
plt.xlabel('Experience (Years)')
plt.ylabel('Salary')

plt.show()
```



Experiment 1.6: Display the top 10 employees with the highest salary.

```
top_10 = df.sort_values('Salary', ascending=False).head(10)

print(top_10)
```

	Employee_ID	Name	Department	Age	Gender	Salary	Experience_Years
87	E088	Diana88	Marketing	36	Male	119000	33
85	E086	Omar86	Marketing	58	Male	119000	24
22	E023	Rahul23	Finance	57	Male	117000	7
14	E015	Ishan15	Operations	57	Male	117000	20
58	E059	Neha59	IT	43	Male	117000	16
48	E049	Charlie49	IT	26	Male	116000	15
64	E065	Bob65	Finance	35	Male	115000	6
55	E056	Sneha56	Operations	41	Male	115000	35
31	E032	George32	Operations	41	Female	115000	23
37	E038	Nisha38	IT	28	Male	114000	27

Experiment 1.7: Find the highest salary in every department.

```
highest_salary = df.groupby('Department')['Salary'].max()

print(highest_salary)
```

```

Department
Finance      117000
HR            113000
IT            117000
Marketing     119000
Operations    117000
Sales         114000
Name: Salary, dtype: int64

```

Experiment 1.8: Display employees whose salary is greater than the overall average salary.

```
average_salary = df['Salary'].mean()

employees = df[df['Salary'] > average_salary]

print(employees)
```

	Employee_ID	Name	Department	Age	Gender	Salary	Experience_Years
1	E002	Diana2	Operations	56	Male	105000	27
3	E004	Neha4	HR	57	Male	113000	34
7	E008	Neha8	Finance	24	Female	98000	7
8	E009	Mohan9	HR	57	Female	110000	23
9	E010	Vikram10	IT	26	Male	114000	14
10	E011	Julia11	HR	36	Male	78000	17
12	E013	George13	Operations	39	Male	107000	10
13	E014	Sneha14	Operations	37	Male	89000	24
14	E015	Ishan15	Operations	57	Male	117000	20
15	E016	Bob16	IT	24	Female	81000	17
18	E019	Hina19	Operations	57	Female	104000	27
21	E022	Nisha22	Sales	26	Female	78000	29
22	E023	Rahul23	Finance	57	Male	117000	7
24	E025	Farah25	Marketing	22	Female	94000	11
26	E027	Farah27	Sales	55	Male	106000	20
29	E030	Charlie30	Sales	30	Male	114000	30
31	E032	George32	Operations	41	Female	115000	23
33	E034	Charlie34	Finance	23	Male	105000	14
37	E038	Nisha38	IT	28	Male	114000	27
40	E041	George41	Sales	50	Male	84000	11
41	E042	Ishan42	Marketing	37	Male	86000	35
44	E045	Bob45	IT	46	Male	79000	16
48	E049	Charlie49	IT	26	Male	116000	15
49	E050	Mohan50	HR	58	Male	104000	2
52	E053	Omar53	Finance	26	Male	88000	6
55	E056	Sneha56	Operations	41	Male	115000	35
58	E059	Neha59	IT	43	Male	117000	16

61	E062	Farah62	Operations	50	Female	101000	0
64	E065	Bob65	Finance	35	Male	115000	6
67	E068	Hina68	Finance	32	Male	78000	2
69	E070	Hina70	IT	23	Male	81000	21
70	E071	Ishan71	HR	39	Female	112000	32
73	E074	Neha74	Marketing	44	Female	85000	32
76	E077	Charlie77	Operations	43	Female	114000	7
78	E079	Julia79	Sales	30	Male	83000	24
79	E080	Farah80	Sales	58	Female	81000	35
80	E081	Alice81	Finance	40	Male	85000	20
81	E082	Omar82	Marketing	50	Male	95000	30
82	E083	Farah83	Operations	27	Female	95000	21
84	E085	Alice85	HR	37	Female	108000	4
85	E086	Omar86	Marketing	58	Male	119000	24
86	E087	Priya87	Marketing	37	Male	113000	0
87	E088	Diana88	Marketing	36	Male	119000	33
89	E090	Ethan90	Marketing	55	Female	86000	32
90	E091	Nisha91	Sales	50	Male	90000	28
91	E092	Ishan92	IT	39	Female	110000	15
95	E096	Nisha96	Finance	56	Female	83000	3
96	E097	George97	Marketing	46	Male	103000	24
97	E098	Priya98	HR	44	Female	79000	26
98	E099	Sneha99	Operations	56	Male	92000	14
99	E100	Ishan100	Marketing	53	Male	79000	21

Experiment 1.9: Calculate the average years of experience in each department.

```
avg_experience = df.groupby('Department')['Experience_Years'].mean()

print(avg_experience)
```

```
Department
Finance      11.166667
HR           19.809524
IT           18.388889
Marketing    21.900000
Operations   16.875000
Sales        23.846154
Name: Experience_Years, dtype: float64
```

Experiment 1.10: Display the distribution of employee ages using a histogram.

```
plt.hist(df['Age'], bins=10)

plt.title('Distribution of Employee Ages')
plt.xlabel('Age')
plt.ylabel('Number of Employees')
```



```
plt.show()
```

